

META ADS CAMPAIGN PERFORMANCE TRACKER

Product Requirements Document

Internal multi-tenant reporting & intelligence platform for Meta (Facebook) Ads — built for real estate marketing operations, architected for white-label expansion.

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1. Executive Summary

This document specifies a purpose-built, internal reporting and analytics platform for monitoring Meta (Facebook) Ads campaign performance across one Business Manager and multiple Ad Accounts, initially serving a real estate marketing team of 10–25 people, and architected from day one as a multi-tenant (workspace-based) system so it can be white-labelled to other clients later.

The system is explicitly a monitoring, analysis, comparison, and reporting tool — it does not create, edit, or manage live ad campaigns. It becomes the single source of truth that answers, at a glance, what management actually asks every day: where money is going, which campaigns are working, which need attention, and what to do next.

The platform is designed to be built in a focused MVP first — core campaign tracking, lead analytics, and role-based dashboards — with advanced analytics (AI-driven recommendations, predictive pacing, lead-quality scoring) deferred to later phases. This phased approach keeps the initial build scoped to what management needs on day one, while the underlying architecture is designed to support the full feature set as it's phased in.

2. Problem Statement

Today, campaign performance data lives natively inside Meta Ads Manager, which is built for media buyers running campaigns — not for management asking business questions. There is no single view that connects ad spend to property, city, manager, or executive performance, and no historical, comparative, or role-specific reporting layer.

Core pain points this system resolves

- No unified view across 300+ campaigns spanning multiple ad accounts under one Business Manager.
- Leadership cannot quickly answer "what changed since yesterday" or "which campaign needs attention now."
- No systematic way to attribute spend and leads to property, city, or responsible manager/executive.
- No standardized daily/weekly/monthly reporting — reporting today is manual and inconsistent.
- No alerting — underperformance (CTR drop, budget exhaustion, learning-limited status) is discovered late, not proactively.
- No performance accountability layer for managers, supervisors, and executives individually.
- Revenue/ROI data is partial (deal closures aren't always tracked against ad spend), so true ROAS is currently unmeasurable — the system needs to support an estimated-ROI model using assumed conversion rate and average deal value until full CRM integration exists.

3. Objectives

Each objective below states the business question it answers and why that matters — this is intentional: every module in this PRD traces back to one of these questions, so nothing is built without a clear business purpose.

Objective	Business Question Answered	Why It Matters
Real-time spend visibility	How much are we spending today, across all accounts?	Prevents budget overruns and surprises at month-end billing.
Profitability tracking	Which campaigns are profitable (or trending toward it)?	Redirects budget toward what works, away from what doesn't.
Account health monitoring	Which ad accounts are underperforming?	Catches account-level issues (policy flags, low delivery) before they compound.

Objective	Business Question Answered	Why It Matters
Attention triage	Which campaign needs immediate attention?	Turns a reactive team into a proactive one — issues are surfaced, not discovered.
People performance	Which manager/executive performs best?	Creates accountability and a fair basis for incentives and coaching.
ROI visibility	What is today's (estimated) ROI?	Connects ad spend to business outcomes, not just vanity ad metrics.
Property intelligence	Which property generated the most leads?	Informs where to invest marketing budget across the portfolio.
Geographic intelligence	Which city performs best?	Guides geographic expansion and city-level budget allocation.
Creative intelligence	Which creative performs best?	Reduces creative fatigue and guides the creative team's next brief.
Audience intelligence	Which audience performs best?	Improves targeting efficiency and lowers cost per lead over time.
Change detection	What changed compared to yesterday?	Day-over-day deltas are often more actionable than absolute numbers.
Budget action	Which campaigns should be paused or get more budget?	Turns analysis into a direct, actionable recommendation.

4. Success Metrics

How we'll know the MVP succeeded, 60–90 days after launch:

- 100% of active campaigns (300+) syncing daily with zero manual data entry.
- Daily/weekly reports generated and delivered automatically to each role, replacing manual reporting.
- Time-to-detect a critical issue (e.g. budget exhausted, CTR collapse) drops from "discovered days later" to under 2 hours via alerts.
- At least 80% of active users (managers, supervisors, executives) log in weekly without being reminded — indicating the dashboards are genuinely useful, not just built.
- Leadership can answer any of the 12 objective questions above in under 60 seconds using the dashboard, without asking a team member to pull a manual report.

5. Assumptions & Constraints

5.1 Assumptions

- One Facebook Business Manager houses multiple Ad Accounts at launch; additional Business Managers (for future white-label clients) will map to additional workspaces.
- Campaign volume at launch is 300+ active campaigns, scaling over time — the data model and sync jobs must be designed for this volume from day one, not retrofitted.

- Lead destination is mixed: some campaigns use native Meta Lead Forms (leads retrievable via Meta API), others send traffic to landing pages/websites (leads arrive via pixel/CRM webhook, not the Meta API directly). The system must support both ingestion paths.
- True revenue/ROAS data is only partially available. Until full CRM integration exists, ROI will be computed using an editable assumed conversion rate and average deal value per property/campaign type, clearly labeled as "Estimated ROI" everywhere it's shown — never presented as confirmed revenue.
- No strict campaign naming convention exists today. The system will ship with a recommended naming convention ([Manager] - [Client/Property] [Objective] Campaign [Month-Year]) that can be auto-parsed to populate property/city/manager tags, plus a manual tagging fallback in-app for campaigns that don't follow it.
- The team size (10–25 people, 1–2 managers) at launch is small enough that some personas (e.g. Supervisor and Manager) may be held by the same person — the persona model below is designed to scale to larger client organizations without forcing extra roles on this team now.

5.2 Constraints

- Subject to Meta Marketing API rate limits and data retention windows (Meta API insight data is generally reliable in detail for ~37 months, with some breakdowns limited further).
- Meta attribution windows and platform-reported metrics won't perfectly reconcile with CRM-reported leads/sales — the system should surface both and be transparent about the gap, not force a false reconciliation.
- WhatsApp Business API (for alerting) requires an approved WhatsApp Business Account and message templates pre-approved by Meta — this is a dependency with lead time, not an instant integration.
- Multi-tenant white-labeling later implies workspace-level data isolation must be correct from day one (not bolted on) — this drives the Postgres Row-Level-Security-based architecture recommended in Section 23.

6. Out of Scope / Non-Goals

Explicitly excluded from this system, to prevent scope creep during development:

- Creating, editing, launching, or pausing live ad campaigns (this is a read-only reporting layer over Meta's Ads Manager, not a replacement for it).
- Creative design or asset production tools (the system displays creative performance, not a creative editor).
- Full CRM functionality (lead assignment, sales pipeline management, calling) — the system consumes lead/CRM data for analytics, it does not replace a CRM like Ashiyana.
- Ad platforms other than Meta (Google Ads, LinkedIn Ads, etc.) at launch — the architecture should not preclude this later, but it is not built now.
- Automated bidding, budget changes, or campaign actions triggered by the system (all recommendations are advisory; a human takes the action in Ads Manager).
- Payment processing / billing for white-label clients — this PRD covers the product itself, not the commercial SaaS billing layer (that would be a separate workstream once white-labeling is prioritized).

7. User Personas

Seven personas cover the full hierarchy this system needs to support — from a solo owner today to a larger client organization (Director, Sales Manager layers) once white-labelled. Not every persona will have a distinct human behind it at launch; smaller teams simply collapse two personas into one user.

7.1 CEO / Business Owner

Purpose: Give the person ultimately accountable for the business a 30-second read on marketing health without needing to ask anyone.

Advantage: Removes the founder as a bottleneck for status updates and replaces gut-feel with a real number every morning.

Attribute	Detail
Goals	Understand overall marketing ROI and spend efficiency at a glance; make go/no-go budget calls.
Responsibilities	Approve major budget shifts; hold managers accountable; review weekly/monthly trends.
Key KPIs Watched	Total spend, estimated ROI, cost per lead, top/bottom performing properties, MoM trend.
Dashboards	CEO Dashboard (Section 11.1)
Permissions	View all workspaces/accounts (full read access), export, approve budget-shift recommendations. No edit access to raw campaign data.
Reports Received	Daily digest (WhatsApp/email), Monthly Executive Summary.

7.2 Marketing Director

Purpose: Give the person owning marketing outcomes end-to-end a way to manage the whole account portfolio, not just one campaign at a time.

Advantage: Lets one person spot cross-account patterns (e.g. a city underperforming everywhere) that would be invisible looking account-by-account in Ads Manager.

Attribute	Detail
Goals	Optimize budget allocation across accounts, properties and managers; set targets.
Responsibilities	Own the daily/weekly review cadence; set and adjust daily/weekly/monthly targets; escalate account-level issues to Meta support when needed.
Key KPIs Watched	ROAS/ROI, budget pacing, account health, manager leaderboard, target achievement %.
Dashboards	Management Dashboard, Marketing Director Dashboard (Section 11.2–11.3)
Permissions	Full read access across all workspaces they're assigned to; edit targets; approve alerts thresholds; export all reports.
Reports Received	Daily digest, Weekly Performance Report, Manager Performance Report.

7.3 Marketing / Campaign Manager

Purpose: Give the person running day-to-day campaign strategy the operational detail needed to act, not just observe.

Advantage: Cuts the time spent manually pulling Ads Manager exports before every status meeting.

Attribute	Detail
Goals	Hit lead/CPL targets for the accounts they own; manage a team of executives.
Responsibilities	Daily campaign monitoring; reassign underperforming campaigns; brief executives on priorities.
Key KPIs Watched	CPL, CTR, budget utilization, learning-limited campaigns, executive-level breakdown.
Dashboards	Manager Dashboard (Section 11.4)
Permissions	Full read/edit on assigned ad accounts and their team's data; cannot see other managers' compensation-linked KPIs unless Director-level.
Reports Received	Daily digest, Weekly Team Performance Report.

7.4 Supervisor

Purpose: Give the person overseeing a cluster of executives a mid-layer view — enough detail to coach, without the account-level noise a Director needs.

Advantage: Bridges the gap between campaign-level detail and people-management, so coaching conversations are backed by data, not memory.

Attribute	Detail
Goals	Ensure executives are actioning alerts and following process; maintain campaign hygiene (naming, tagging).
Responsibilities	Daily campaign QA; first responder to alerts before escalating to Manager.
Key KPIs Watched	Alert response time, campaign tagging completeness, team CPL trend.
Dashboards	Supervisor Dashboard (Section 11.5)
Permissions	Read access to assigned executives' campaigns; can acknowledge/resolve alerts; cannot change budgets.
Reports Received	Daily digest, Weekly Alert Summary.

7.5 Performance Marketing Executive

Purpose: Give the person closest to individual campaigns a focused, low-noise view of just their own campaigns.

Advantage: Prevents information overload — an executive managing 20 campaigns shouldn't have to wade through a 300-campaign dashboard to find their own numbers.

Attribute	Detail
Goals	Keep assigned campaigns healthy; respond to alerts quickly; hit individual CPL targets.
Responsibilities	Daily monitoring of assigned campaigns; tagging property/city correctly; flagging creative fatigue.
Key KPIs Watched	Own CPL, CTR, frequency, budget pacing on assigned campaigns only.
Dashboards	Campaign Executive Dashboard (Section 11.6)
Permissions	Read access limited to assigned campaigns only; can acknowledge alerts; no export of other executives' data.
Reports Received	Daily digest (own campaigns only).

7.6 Data Analyst / Growth Analyst

Purpose: Give the person doing deeper analysis (cohort trends, audience/creative testing) unrestricted access to filter, slice, and export.

Advantage: Avoids analysts having to request raw exports from engineering every time a new question comes up.

Attribute	Detail
Goals	Identify trends across properties, audiences, and creatives that inform strategy; build ad-hoc comparisons.
Responsibilities	Build custom reports; run comparison views; support the Director with data for planning.
Key KPIs Watched	All KPIs, at full historical depth, with drill-down and pivot access.
Dashboards	All dashboards (read-only across workspace), Export Centre, Report Builder.
Permissions	Full read + export access across the workspace; no edit access to live data or settings.
Reports Received	On-demand — this persona pulls, rather than receives, reports.

7.7 Administrator (Workspace Admin)

Purpose: Give one person the ability to configure the system itself — users, roles, integrations — without needing engineering involvement for routine changes.

Advantage: Makes the system self-serve for onboarding a new workspace/client, which is essential once white-labeling begins.

Attribute	Detail
Goals	Keep the system correctly configured, secure, and connected to the right Meta assets.
Responsibilities	Manage users and roles; connect/reconnect Meta Business Manager and ad accounts; configure alert thresholds and notification channels; review audit logs.
Key KPIs Watched	System health metrics: sync success rate, API error rate, active user count.
Dashboards	Admin Panel (Section 20).
Permissions	Full system access: user management, role management, integration settings, audit log access. No implicit access to see business KPIs unless also assigned a business persona.
Reports Received	System Health Report (weekly).

8. User Stories

A representative set per persona — the full backlog will expand these during sprint planning, but every functional requirement in Section 9 traces back to one of these.

Persona	User Story
CEO	As a CEO, I want a single number for today's estimated ROI across all accounts, so I don't have to ask anyone for a status update.
Marketing Director	As a Marketing Director, I want to see which of our cities is underperforming this week vs last week, so I can reallocate budget before month-end.
Marketing Manager	As a Manager, I want to see which of my team's campaigns are Learning Limited, so I can fix them before they waste further spend.
Supervisor	As a Supervisor, I want a list of unacknowledged alerts older than 2 hours, so I can chase the right executive.
Executive	As an Executive, I want to see only my own campaigns with today's spend and leads, so I'm not distracted by the full account list.
Data Analyst	As an Analyst, I want to compare creative performance across all properties in a date range, so I can recommend what to brief next.
Administrator	As an Admin, I want to add a new Ad Account to an existing workspace in under 5 minutes, so onboarding a new client doesn't need engineering.

9. Functional Requirements — Modules

Every module below states its Purpose (why we're building it) and Advantage (what it unlocks), followed by concrete requirements, and is tagged MVP or Phase 2 per the phasing decision in Section 24.

9.1 Dashboard (role-based landing page) [MVP]

Purpose: Give every role a single first screen that answers their top 3 questions without clicking anywhere.

Advantage: Reduces time-to-insight from minutes of manual filtering to seconds on login.

- Role-aware landing page — content differs by persona (Section 11).
- Today vs Yesterday delta on headline KPIs (spend, leads, CPL).
- Top 3 alerts requiring attention, pinned at the top.
- Quick filters: date range, account, property, manager.

9.2 Campaign Monitoring [MVP]

Purpose: Give a live, sortable list of every campaign with health status at a glance.

Advantage: Replaces manually scrolling Ads Manager's 300+ campaign list with a sortable, filterable, taggable table.

- Table view: campaign name, account, status, objective, spend, results, CPL/CPA, delivery status, learning status.
- Health indicator (Green/Amber/Red) computed from CTR, frequency, and pacing thresholds.
- Bulk tagging: assign property, city, manager, executive to campaigns lacking a naming-convention match.
- Search and multi-filter combination (see Section 14).

9.3 Campaign Performance (detail view) [MVP]

Purpose: Give a single campaign's full story — trend, ad sets, ads, creatives — in one drill-down screen.

Advantage: Removes the need to jump between Ads Manager tabs to understand one campaign's full picture.

- Time-series trend chart for spend, CTR, CPL over the selected date range.
- Ad set and ad-level breakdown table.
- Creative thumbnails with performance overlay.
- Day-over-day and week-over-week comparison toggle.

9.4 Lead Analytics [MVP]

Purpose: Connect ad spend to the leads it actually produced, regardless of whether they came via Meta Lead Forms or a landing page.

Advantage: This is the single most requested business answer ("which campaign generated maximum leads") — without it, spend and outcome stay disconnected.

- Unified lead table merging Meta Lead Ads data and landing-page/CRM-sourced leads (dual ingestion per Section 5.1).
- Lead volume trend by campaign, property, city, source.
- Cost per Lead (CPL) computed per campaign/property/city/manager.
- Lead quality tagging field (Phase 2: auto-scored; MVP: manual tag synced from CRM where available).

9.5 Property Analytics [MVP]

Purpose: Roll campaign performance up to the property level — the unit real estate leadership actually thinks in.

Advantage: Lets leadership answer "which property generated maximum leads" directly, instead of mentally aggregating campaign names.

- Property-level summary: total spend, leads, CPL, estimated ROI.
- Property comparison view (side-by-side, up to 5 properties).
- Property leaderboard, sortable by any KPI.

9.6 Audience Analytics [Phase 2]

Purpose: Show which audience segments are actually converting, not just which have the most reach.

Advantage: Improves targeting efficiency over time by making audience fatigue and winners visible, not buried in Ads Manager breakdowns.

- Audience performance table: age, gender, placement, device breakdowns.
- Audience overlap / fatigue indicator.
- Best-performing audience recommendation per property type.

9.7 Creative Analytics [Phase 2]

Purpose: Show which ad creative (image/video/copy) is actually driving results, and flag fatigue before it hurts CPL.

Advantage: Gives the creative team a data-backed brief for what to make next, instead of guessing.

- Creative-level performance table with thumbnails: CTR, hook rate, cost per view, CPL.
- Creative fatigue alert (frequency rising + CTR declining).
- Creative leaderboard by property/campaign objective.

9.8 Budget Tracking & Pacing [MVP]

Purpose: Show, in real time, whether each campaign/account is on pace to over- or under-spend its budget.

Advantage: Prevents the two most common budget failures: running out mid-month, or under-spending an approved budget.

- Budget utilization % and pacing indicator (ahead/on-track/behind) per campaign and account.
- Forecast spend (simple linear projection at MVP; Phase 2: seasonality-aware forecast).
- Remaining budget and days-until-exhaustion estimate.

9.9 Daily / Weekly / Monthly Reports [MVP]

Purpose: Automatically generate the reports people currently build by hand.

Advantage: Frees the team from manual report-building and guarantees consistent, comparable reporting every period.

- Auto-generated Daily Digest (per role) delivered via email + WhatsApp + in-app.
- Weekly Performance Report (Monday morning, prior week summary).
- Monthly Executive Summary (first business day of the month).
- Phase 2: Quarterly/Yearly and fully custom date-range reports, scheduled report builder.

9.10 Team & Employee Performance [MVP]

Purpose: Turn individual manager/supervisor/executive performance into a comparable, visible leaderboard.

Advantage: *Creates fair, data-backed accountability instead of anecdotal performance reviews.*

- Leaderboard by manager and by executive: CPL, lead volume, alert response time.
- Individual performance trend view (for 1:1 coaching conversations).

9.11 Alerts & Notifications [MVP]

Purpose: *Proactively surface problems the moment they cross a defined threshold, instead of waiting for someone to notice.*

Advantage: *Converts the system from a passive report into an active early-warning system — this is the highest-leverage module in the whole platform.*

- Rule engine per Section 17, configurable thresholds per workspace.
- Delivery via Email, WhatsApp Business API, and in-app push (PWA) as decided in Section 5.
- Acknowledge / Resolve / Escalate workflow with audit trail.

9.12 Export Centre [MVP]

Purpose: *Let any authorized user get their data out, in the format they need, without asking engineering.*

Advantage: *Removes a common support-ticket category ("can you send me this as Excel") entirely.*

- Export to CSV/XLSX for any table view, respecting current filters.
- PDF export for report views (branded, workspace-logo aware for white-label).
- Export history/audit log.

9.13 Admin & Role Management [MVP]

Purpose: *Let an Administrator manage users, roles, and permissions without engineering involvement.*

Advantage: *Essential for onboarding new workspaces quickly once white-labeling starts.*

- User invite/deactivate flow.
- Role assignment per the RBAC matrix (Section 21).
- Workspace-level settings: branding (white-label), timezone, currency, alert thresholds.

9.14 Meta API Integration [MVP]

Purpose: *Reliably pull campaign, ad set, ad, and insight data from Meta's Marketing API on a schedule.*

Advantage: *This is the data foundation everything else depends on — without reliable sync, every dashboard is stale or wrong.*

- System User-based authentication per workspace (a dedicated Business Manager System User with Employee-level access, not a personal login).
- Daily scheduled sync (cron) plus on-demand manual refresh.
- Sync status visibility (last synced timestamp, error state) surfaced to Admins.
- Graceful handling of Meta API rate limits, token expiry, and partial failures (retry with backoff).

9.15 Audit Logs [MVP]

Purpose: *Record who did what, when — every settings change, alert resolution, and export.*

Advantage: *Required for accountability once multiple people and multiple client workspaces share the system, and for diagnosing "who changed this threshold" disputes.*

- Immutable log of user actions: logins, exports, settings changes, alert acknowledgments, role changes.
- Admin-only searchable log view with date/user/action filters.

9.16 Settings / Workspace Management [MVP]

Purpose: *Let each workspace (client) configure its own accounts, users, and preferences independently.*

Advantage: *This is what makes the multi-tenant architecture actually usable by non-technical admins, not just correct in the database.*

- Workspace switcher for users who belong to multiple workspaces (e.g. the founder across real estate + future clients).
- Per-workspace: connected ad accounts, currency, timezone, branding, alert thresholds, notification channel preferences.

10. Non-Functional Requirements

Category	Requirement
Performance	Dashboard initial load under 2s for cached data; filtered queries under 3s at 300+ campaign scale.
Scalability	Data model and sync jobs must handle 10x campaign volume (3,000+) and multiple workspaces without redesign.
Caching	Aggregate KPI rollups pre-computed on sync (not calculated live on every dashboard load).
Offline support (PWA)	Last-synced dashboard data viewable offline; clear "stale data" indicator when offline.
Accessibility	WCAG 2.1 AA color contrast on all charts and KPI cards; keyboard-navigable tables.
Security	Row-Level Security enforced at the database layer per workspace (not just application-layer checks); encrypted secrets for Meta API tokens.
Auditability	Every settings change and data export logged (Section 9.15).
Availability	Target 99.5% uptime for the dashboard; sync jobs must alert Admins on failure, not fail silently.
Error handling	User-facing errors are plain-language; technical detail logged, not shown to end users.
Monitoring	Sync job success/failure, API error rates, and page load times tracked and visible to Admins.

11. Dashboard Specifications

Five role-based dashboards cover the seven personas (Supervisor and Executive share a template with scope-based filtering; Manager and Director dashboards differ mainly in breadth, not structure).

11.1 CEO Dashboard

Purpose: Answer "is the business's marketing healthy" in one screen, in business language, not ad jargon.

Advantage: The only dashboard the CEO ever needs to open — no training required to read it.

Widgets / Cards: Total spend (MTD/today), Estimated ROI, Estimated Revenue, Top 3 / Bottom 3 properties, Spend trend line (30-day), Alert summary panel, Manager leaderboard (top-line only).

Filters & Drill-down: Date range, workspace (if multi-client access). No campaign-level drill-down — links out to Management Dashboard for detail.

11.2 Management / Marketing Director Dashboard

Purpose: Give the person who owns marketing outcomes the full portfolio view with drill-down into any dimension.

Advantage: Lets one person manage 300+ campaigns' worth of complexity without needing to open Ads Manager.

Widgets / Cards: Spend & leads trend (with day-over-day delta), Account health heatmap, Property leaderboard, City leaderboard, Manager leaderboard, Budget pacing table, Alert panel, Target achievement gauge (daily/weekly/monthly).

Filters & Drill-down: Date range, account, property, city, manager. Drill-down from any chart into Campaign Monitoring.

11.3 Manager Dashboard

Purpose: Give a manager everything needed to run their assigned accounts day-to-day.

Advantage: Turns the manager's morning routine from a manual Ads Manager sweep into a 5-minute dashboard review.

Widgets / Cards: My accounts' spend/leads/CPL today vs yesterday, Campaigns needing attention (Red/Amber), My team's leaderboard, Budget pacing for my accounts, Alert panel (my scope only).

Filters & Drill-down: Date range, account, executive, property. Drill-down to Campaign Performance detail.

11.4 Supervisor Dashboard

Purpose: Give a supervisor visibility into their executives' alert responsiveness and campaign hygiene.

Advantage: Makes coaching data-driven — a supervisor can point to a specific unactioned alert, not a vague impression.

Widgets / Cards: Unacknowledged alerts (my team), Campaign tagging completeness %, Team CPL trend, Executive comparison table.

Filters & Drill-down: Date range, executive. Drill-down to individual executive's campaigns.

11.5 Campaign Executive Dashboard

Purpose: Give an executive a focused view of only their own campaigns.

Advantage: Removes noise — an executive should never have to filter out 280 campaigns that aren't theirs just to find their own 20.

Widgets / Cards: My campaigns list with health indicator, My CPL/CTR/frequency trend, My open alerts, My leads today.

Filters & Drill-down: Date range only (scope is pre-filtered to assigned campaigns).

11.6 Analyst / Data Analyst View

Purpose: *Give the analyst persona unrestricted pivot and comparison tools across the full workspace.*

Advantage: *Supports ad-hoc questions leadership asks that don't fit a pre-built dashboard.*

Widgets / Cards: Pivot table builder, multi-property/multi-campaign comparison view, full KPI library access, saved custom views.

Filters & Drill-down: All filters in Section 14 available in combination; no scope restriction within the workspace.

12. Screen-by-Screen Requirements

Screen	Key Requirements
Login / Workspace Switcher	Email/password or SSO login; if user belongs to >1 workspace, workspace picker shown before dashboard.
Dashboard (role-based)	As specified per persona in Section 11; loads in under 2s from cache.
Campaign Monitoring (list)	Sortable/filterable table, health indicators, bulk tagging, saved filter views.
Campaign Detail	Trend charts, ad set/ad breakdown, creative thumbnails, comparison toggle.
Lead Analytics	Unified lead table, source breakdown, CPL by dimension, quality tag field.
Property Analytics	Property summary cards, comparison view, leaderboard.
Reports Centre	List of generated reports (daily/weekly/monthly), download/resend, Phase 2 custom report builder.
Alerts Centre	Full alert feed, filter by severity/status, acknowledge/resolve/escalate actions.
Export Centre	Export history, format selector, filter-aware export trigger.
Admin Panel	User management, role assignment, workspace settings, connected ad accounts, audit log.

13. KPI Framework

KPIs are grouped by category rather than listed flat — this makes the (very long) list from the original brief usable in a UI, since each group maps naturally to a filter tab or dashboard section.

13.1 Delivery & Reach

KPI	Definition / Note
Spend	Total amount spent, by campaign/account/day.
Budget & Budget Utilization %	Configured budget vs. amount spent so far.
Impressions	Number of times an ad was shown.
Reach	Number of unique people who saw an ad.
Frequency	$\text{Impressions} \div \text{Reach}$ — high frequency signals ad fatigue.
CPM	Cost per 1,000 impressions.
Delivery Status / Learning Limited	Meta's own delivery health signal, surfaced directly, not recalculated.

13.2 Engagement & Clicks

KPI	Definition / Note
CTR / Unique CTR	Click-through rate, overall and per unique user.
CPC	Cost per click.
Outbound Click / Link Click	Clicks that leave the platform vs. any link click.
Landing Page View	Confirms the click resulted in an actual page load.
Video View / View Rate / Hook Rate	Video-specific engagement depth, key for creative diagnostics.
Post Engagement (Likes, Comments, Shares, Saves)	Organic-style engagement signals on the ad unit.

13.3 Cost Efficiency

KPI	Definition / Note
Cost per Result	Meta's own optimization-event cost metric.
Cost per View / Cost per Purchase	Cost efficiency for video and purchase-objective campaigns.

13.4 Lead & Conversion (core to real estate)

KPI	Definition / Note
Lead / Qualified Lead	Raw lead count vs. CRM-confirmed qualified lead.
CPL (Cost per Lead)	$\text{Spend} \div \text{Leads}$ — the single most-watched KPI for this business.
Conversion Rate	$\text{Leads} \div \text{Clicks}$, or $\text{Qualified Leads} \div \text{Leads}$ depending on the funnel stage viewed.
Lead Quality Score	MVP: manual tag from CRM feedback. Phase 2: modeled score from historical conversion patterns.

KPI	Definition / Note
Estimated ROAS / Estimated ROI	Computed using assumed conversion % and average deal value per property (Section 5.1) — always labeled "Estimated".

13.5 Budget & Pacing

KPI	Definition / Note
Pacing (Ahead / On-track / Behind)	Spend-to-date vs. expected spend-to-date given the budget period.
Forecast Spend / Forecast Leads	Linear projection at MVP; seasonality-aware in Phase 2.
Remaining Budget	Budget minus spend-to-date.
Daily / Weekly / Monthly Target & Achievement %	User-configured targets vs. actuals.

13.6 Account & Ranking Signals

KPI	Definition / Note
Quality / Engagement / Conversion Ranking	Meta's relative ad-quality diagnostics, surfaced as-is.
Auction Overlap	Flags when your own campaigns compete against each other.

Phase 2 note: Lead Quality Score and full ROAS become precise once CRM integration is deeper (Section 24); until then they are clearly marked as estimates everywhere they appear in the UI.

14. Filters

Filters are combinable (AND logic across groups, OR within a group — e.g. City: Pune OR Mumbai, AND Manager: X).

Filter Group	Options
Date	Today, Yesterday, Last 7 Days, Last 30 Days, Custom Range, Week-over-Week / Month-over-Month compare toggle.
Campaign	Campaign, Objective, Status, Buying Type.
Account / Structure	Ad Account, Business Manager (Phase 2, once multi-BM), Platform, Placement, Device.
Geography	Country, State, City.
Business Attribution	Property, Sales Team, Marketing Manager, Campaign Manager, Supervisor, Executive.
Audience & Creative	Audience, Creative, Gender, Age.
Attribution / Tech	Lead Source, Pixel, Event, UTM parameters.

15. Reports

Report	Frequency	Recipients	Phase
Daily Digest	Daily, 9 AM IST	All roles (scoped to their access)	MVP
Weekly Performance Report	Every Monday	Director, Manager, Supervisor	MVP
Monthly Executive Summary	1st business day of month	CEO, Director	MVP
Manager Performance Report	Weekly	Director	MVP
Property Performance Report	Weekly	Director, CEO	MVP
Campaign Performance Report	On-demand	Manager, Supervisor, Executive	MVP
Creative Performance Report	Weekly	Manager, Analyst	Phase 2
Audience Performance Report	Weekly	Analyst	Phase 2
Lead Quality / Sales Funnel Report	Weekly	Director, Manager	Phase 2 (needs deeper CRM link)
Quarterly / Yearly / Custom Report Builder	On-demand	All (permission-gated)	Phase 2

16. Visualizations

Chart choice mapped to what each metric type needs to communicate — not chosen for visual variety.

Metric Type	Recommended Chart
Single headline number (spend, ROI today)	KPI Card with delta indicator (▲/▼vs. yesterday).
Trend over time (spend, CPL, leads)	Line Chart / Area Chart.
Comparison across categories (properties, cities, managers)	Bar Chart / Leaderboard.
Composition (spend by objective, leads by source)	Donut Chart (avoid Pie for >4 segments).
Hierarchical composition (spend by account > campaign > ad set)	Treemap.
Two-dimensional health view (campaign x day)	Heatmap.
Geographic performance (city/state)	Geo Map.
Relationship between two metrics (spend vs. leads)	Scatter Plot / Bubble Chart (bubble size = 3rd metric, e.g. CPL).
Funnel-shaped drop-off (impressions > clicks > leads > sales)	Funnel Chart.
Contribution to a total change (why spend changed WoW)	Waterfall Chart.
Target vs. actual (daily/weekly/monthly target)	Gauge Chart.
Ranked list (top managers, top properties)	Leaderboard (styled table, not a generic chart).
Multi-dimensional cross-tab (Analyst use)	Pivot Table / Matrix.

17. Alerts & Notification Rules

Alerts are the highest-leverage module (Section 9.11) — this table is the initial rule set; all thresholds are configurable per workspace by an Admin.

Alert	Trigger Condition	Default Severity	Channel
CTR below threshold	CTR < workspace-defined minimum for 2+ consecutive days	Amber	Email + In-app
CPA / CPL increased	CPL rises >25% vs. trailing 7-day average	Amber	Email + In-app
Budget exhausted	Remaining budget = 0 or campaign paused by Meta for budget	Red	WhatsApp + Email + In-app
Campaign rejected	Meta returns ad review rejection status	Red	WhatsApp + Email + In-app
Campaign stopped unexpectedly	Active campaign transitions to Paused/Inactive without a logged reason	Red	WhatsApp + In-app
Learning Limited	Meta reports Learning Limited delivery status	Amber	In-app
Frequency high	Frequency exceeds workspace threshold (e.g. >3.5 in 7 days)	Amber	In-app
Estimated ROAS dropped	Estimated ROI trend declines >20% week-over-week	Amber	Email + In-app
Spend anomaly	Day's spend deviates >40% from trailing 7-day average (either direction)	Red	WhatsApp + Email + In-app
Lead volume dropped	Daily leads fall >30% vs. trailing 7-day average	Amber	Email + In-app
Pixel inactive	No pixel events received for 24+ hours on an active campaign	Red	Email + In-app
Sync / API failure	Meta API sync job fails or token expires	Red (Admin-only)	Email + In-app (Admin)

18. Data Model & ER Diagram

Purpose: Define every entity the system tracks and how they relate, before any table is built.

Advantage: Prevents costly schema rework later — entity relationships defined up front catch data-isolation and migration-ordering issues before they're expensive to fix.

18.1 Core Entities

- Workspace (Client) — the multi-tenancy boundary; every other entity below belongs to exactly one Workspace.
- Business Manager — one per Workspace at launch.
- Ad Account — belongs to a Business Manager; multiple per Workspace.
- Campaign — belongs to an Ad Account; tagged with Property, City, Manager, Executive.
- Ad Set — belongs to a Campaign.
- Ad — belongs to an Ad Set; references a Creative.
- Creative — reusable image/video/copy asset, referenced by one or more Ads.
- Audience — targeting definition, referenced by one or more Ad Sets.
- Pixel — tracking asset linked to an Ad Account, referenced by landing-page-sourced Leads.
- Lead — sourced from Meta Lead Ads or Pixel/CRM webhook; linked to a Campaign and Property.
- Property — the real estate unit leads are attributed to.
- Sales Team / Employee — org structure: Manager, Supervisor, Executive, linked to Campaigns and Leads.
- Customer — optional, populated when CRM integration provides post-lead sales data.
- Daily Metrics — one row per Campaign per day: spend, impressions, clicks, leads, etc. (the core fact table).
- Historical Metrics — same shape as Daily Metrics, retained long-term for trend/YoY analysis.
- Alert — generated instance of a triggered rule from Section 17, linked to a Campaign/Account.
- Notification — delivery record of an Alert or Report to a User via a Channel.
- Audit Log — immutable record of a User action.
- Role / Permission — RBAC definitions (Section 21), linked to Users within a Workspace.

18.2 ER Diagram — Provision

An entity-relationship diagram should be produced as a visual artifact alongside this PRD (not embedded as static text) so it stays easy to update as the schema evolves during sprint planning. The core relationships to diagram are:

- Workspace 1—* Business Manager 1—* Ad Account 1—* Campaign 1—* Ad Set 1—* Ad *—1 Creative
- Campaign *—1 Property, Campaign *—1 Manager/Supervisor/Executive (via Sales Team)
- Campaign 1—* Daily Metrics (time-series fact table)
- Campaign 1—* Lead, Lead *—1 Property, Lead *—0..1 Customer (once CRM-linked)
- Ad Account 1—* Pixel, Pixel 1—* Lead (for landing-page-sourced leads)
- Campaign/Account 1—* Alert, Alert 1—* Notification, Notification *—1 User
- Workspace 1—* User, User *—1 Role, Role 1—* Permission

This diagram provision is intentionally left as a to-be-rendered artifact rather than an ASCII sketch in the Word document — say the word and I'll generate the actual ER diagram as a visual right after this PRD is finalized.

19. Database Design

Aspect	Recommendation
Primary/Foreign Keys	UUID primary keys throughout (matches Supabase convention); every child table carries a workspace_id foreign key, even where it could be inferred via a join — this is what makes Row-Level Security simple and fast.
Indexes	Composite index on (workspace_id, campaign_id, date) for Daily Metrics — this is the hot query path for every dashboard.
Partitioning	Partition Daily Metrics / Historical Metrics by month once volume grows past ~6-12 months of 300+ campaigns/day — not needed at MVP launch, but the schema should be partition-ready.
Archiving Strategy	Roll data older than 13 months from Daily Metrics into a compressed Historical Metrics table on a monthly job; dashboards query Historical Metrics transparently for older ranges.
Historical Data Strategy	Preserve raw daily granularity for 24 months minimum (real estate sales cycles are long, per background context); aggregate-only beyond that.
Multi-tenancy Enforcement	Postgres Row-Level Security policy on every table keyed to workspace_id, enforced at the database layer using a SECURITY DEFINER bootstrapping function for initial workspace/user setup, so a bug in application code cannot leak one workspace's data into another's.

20. API Design

Internal REST API (consumed by the Next.js frontend and any future white-label client apps), all routes scoped under `/api/workspaces/{workspaceId}/...` to keep the multi-tenancy boundary explicit at the URL level.

Domain	Representative Endpoints
Dashboard	GET <code>/dashboard/summary</code> , GET <code>/dashboard/alerts</code>
Campaign	GET <code>/campaigns</code> , GET <code>/campaigns/{id}</code> , PATCH <code>/campaigns/{id}/tags</code>
Metrics	GET <code>/metrics/daily</code> , GET <code>/metrics/compare</code>
Reports	GET <code>/reports</code> , GET <code>/reports/{id}</code> , POST <code>/reports/generate</code>
Filters	GET <code>/filters/options</code> (returns valid filter values for current workspace)
Export	POST <code>/export</code> , GET <code>/export/history</code>
Authentication	POST <code>/auth/login</code> , POST <code>/auth/refresh</code> , POST <code>/auth/logout</code>
Notification	GET <code>/notifications</code> , PATCH <code>/notifications/{id}/read</code>
Alert	GET <code>/alerts</code> , PATCH <code>/alerts/{id}/acknowledge</code> , PATCH <code>/alerts/{id}/resolve</code>
Settings	GET/PATCH <code>/workspace/settings</code> , GET/POST <code>/workspace/users</code> , GET/POST <code>/workspace/ad-accounts</code>

21. Role-Based Access Control (RBAC) Matrix

Role	View	Edit (tags/targets)	Delete	Export	Approve	Manage Users/Settings
CEO / Business Owner	All (own workspace)	No	No	Yes	Yes (budget shifts)	No
Marketing Director	All (own workspace)	Yes	No	Yes	Yes	No
Marketing Manager	Assigned accounts	Yes	No	Yes	No	No
Supervisor	Assigned executives' campaigns	Alerts only	No	Limited	No	No
Campaign Executive	Own campaigns only	Tags on own campaigns	No	No	No	No
Data Analyst	All (own workspace)	No	No	Yes	No	No
Administrator	All (system-wide config)	Settings only	Users only	Yes	No	Yes

22. UI/UX Guidelines

Visual and interaction reference points: Power BI and Looker Studio for dashboard density and drill-down patterns; Grafana for alert-panel styling; Meta Ads Manager for table/filter conventions users already know; HubSpot for the overall approachable, non-technical tone appropriate for CEO/Director personas.

- Minimal, professional visual language — data density without clutter; generous whitespace around KPI cards.
- Responsive: full desktop dashboard experience, plus a genuinely usable mobile/PWA view for on-the-go checks (this is a PWA by requirement, not an afterthought).
- Fast: skeleton loaders, not blank screens, while data loads; cached last-known values shown instantly, live data streamed in after.
- Dark/Light mode, respecting system preference by default.
- Consistent health-status color coding across the entire app: Green = healthy, Amber = needs attention, Red = critical — used identically on every screen, not redefined per module.
- White-label readiness: logo, primary color, and workspace name are theme variables, not hardcoded, from the first UI component built.

23. Technical Architecture

Purpose: Recommend a stack that's fast to build, well suited to this scale, and ready for multi-tenancy from day one.

Advantage: Choosing a proven, well-documented stack up front reduces build risk and keeps the team from re-deciding architecture mid-project.

Layer	Recommendation	Rationale
Frontend	Next.js 14 (App Router), TypeScript, Tailwind CSS	Strong PWA support out of the box, and well suited to the dashboard-heavy, data-dense UI this system needs.
Backend	Next.js API routes / server actions (modular monolith, not microservices)	One business-logic layer beats premature microservice complexity at this scale (300+ campaigns, 10-25 users).
Database	Supabase (managed Postgres) with Row-Level Security per workspace	Managed Postgres with built-in Auth and Realtime reduces infrastructure overhead for a small team, while RLS enforces the multi-tenancy boundary at the database layer.
Caching	Pre-computed daily rollup tables (populated by the sync job) + Next.js data caching	Avoids recalculating aggregates live on every dashboard load at 300+ campaign scale.
Authentication	Supabase Auth, workspace-scoped sessions with a workspace switcher	Consistent with the multi-tenant workspace model chosen in Section 5.1/9.16.
Meta Marketing API Integration	System User per workspace (Employee role), scheduled via Vercel Cron	A dedicated System User avoids tying API access to any one person's personal Facebook login, and survives staff turnover.
Background Jobs / Queue	Vercel Cron for scheduled sync; a lightweight queue (e.g. Supabase Edge Functions or a managed queue) for alert evaluation and notification dispatch	Keeps sync and alerting decoupled from user-facing request cycles.
Notifications	Email via a transactional provider (e.g. Resend/SendGrid); WhatsApp via WhatsApp Business API (Meta Cloud API) with pre-approved templates; in-app via Supabase Realtime	Matches the three channels selected in Section 5.
Hosting	Vercel (frontend + API routes)	Zero-config deployment and preview environments, minimal DevOps overhead for a small team.
CI/CD	GitHub Actions → Vercel preview deployments → production on merge to main	Standard, low-maintenance pipeline appropriate for a solo/small-team build.

24. MVP vs. Phase 2 Roadmap

Per your direction: MVP covers core tracking and role-based dashboards; advanced analytics and AI-driven recommendations are deferred.

Phase	Scope	Excludes
MVP (Phase 1)	Dashboard, Campaign Monitoring, Campaign Performance, Lead Analytics, Property Analytics, Budget Tracking, Daily/Weekly/Monthly Reports (auto-generated only), Team Performance, Alerts & Notifications, Export Centre, Admin & Role Management, Meta API Integration, Audit Logs, Workspace Settings.	Audience Analytics, Creative Analytics, AI recommendations, custom report builder, quarterly/yearly reports, deep CRM/lead-quality scoring.
Phase 2	Audience Analytics, Creative Analytics, Lead Quality scoring (model-based), custom/scheduled Report Builder, seasonality-aware forecasting, deeper CRM integration for true (not estimated) ROI.	—
Phase 3 (Future)	Multi-Business-Manager support for onboarding external white-label clients, billing layer, additional ad platforms (Google Ads), native mobile apps.	—

25. Rollout / Migration Plan

- Stage 1 — Connect one Ad Account first (pilot), validate sync accuracy against Ads Manager's own reported numbers before onboarding the remaining accounts.
- Stage 2 — Onboard all remaining Ad Accounts under the single Business Manager; run the system in parallel with any existing manual reporting for one full week to build trust in the numbers.
- Stage 3 — Switch daily/weekly reporting fully to the automated system; retire manual reports.
- Stage 4 — Roll out role-based dashboards to Managers/Supervisors/Executives with a short walkthrough (no formal training module needed at this team size).
- Stage 5 (later) — When the first white-label client is signed, use that onboarding as the test of the multi-tenant workspace model end-to-end before selling it as a repeatable process.

26. Risks

Risk	Impact	Mitigation
Meta API rate limits / breaking changes	Sync failures, stale dashboards	Backoff/retry logic, sync-status monitoring surfaced to Admins, version-pin API calls.
"Estimated ROI" mistaken for confirmed revenue	Bad budget decisions made on unreliable numbers	Persistent "Estimated" labeling and a visible assumptions panel (conversion %, deal value) wherever ROI is shown.
Untagged/inconsistently named campaigns	Broken property/city/manager attribution	Manual tagging fallback (Section 9.2) plus a data-quality completeness indicator on the dashboard.
WhatsApp Business API approval delay	Alert channel unavailable at launch	Launch alerts on Email + In-app first; add WhatsApp once templates are approved, without blocking MVP.
Multi-tenant data leakage across workspaces	Severe — client data exposure once white-labelled	Database-layer Row-Level Security (Section 19), never relying on application-layer checks alone.
Scope creep back toward the full 25-section spec at once	Delayed MVP launch	Phase gating (Section 24) enforced in sprint planning; Phase 2/3 items explicitly deferred, not silently absorbed.

27. Glossary

Term	Definition
CPL	Cost per Lead — total spend divided by number of leads generated.
CPA	Cost per Action/Acquisition — cost per defined conversion event.
CTR	Click-Through Rate — clicks divided by impressions.
CPM	Cost per Mille — cost per 1,000 impressions.
ROAS	Return on Ad Spend — revenue generated divided by ad spend.
ROI	Return on Investment — net profit relative to total investment, broader than ROAS.
Hook Rate	Percentage of viewers who watch the first few seconds of a video ad — an early-engagement signal.
Frequency	Average number of times a unique person has seen an ad within a period.
Learning Limited	A Meta delivery status indicating a campaign lacks enough conversion events to optimize effectively.
Pacing	Whether spend-to-date is ahead of, on track with, or behind the expected rate given the budget period.
Workspace	The multi-tenancy boundary in this system — one client/organization's isolated data and users.
System User	A Meta Business Manager identity used for server-to-server API access, not tied to a personal login.
RLS (Row-Level Security)	A database-layer security mechanism restricting which rows a query can return based on the requesting user's context (here, their workspace).

28. Acceptance Criteria (MVP)

- All active campaigns (300+) sync daily with a visible last-synced timestamp; sync failures generate an Admin alert within 15 minutes.
- Every persona's dashboard (Section 11) loads with role-correct data scope, verified against the RBAC matrix (Section 21) — no user sees data outside their permission scope.
- Daily Digest is delivered automatically via Email and In-app (WhatsApp once approved) to every active user without manual triggering.
- CPL, spend, and lead counts on the dashboard reconcile with Meta Ads Manager's own reported numbers within an agreed tolerance (e.g. $\pm 2\%$) during the pilot stage (Section 25, Stage 1).
- Estimated ROI is visibly and consistently labeled "Estimated" everywhere it appears, with the underlying assumption (conversion %, deal value) visible on hover/click.
- An Administrator can invite a user, assign a role, and connect a new Ad Account without engineering assistance.
- All 12 alert rules in Section 17 fire correctly against test data and can be acknowledged/resolved by an authorized role.
- Row-Level Security is verified to block cross-workspace data access even via a direct API call (not just hidden in the UI).

29. Appendix

29.1 Key Design Decisions Log

A quick-reference summary of the foundational decisions this PRD is built on, for anyone joining the project later:

Decision	Chosen Direction
System scope	New, standalone system — not an extension of any prior project.
Ad account structure	One Business Manager, multiple Ad Accounts at launch.
Client scope	Real estate only at launch; architected to be white-labelled to other clients/verticals later.
Tenancy model	Multi-tenant (workspace-based) from day one, not retrofitted later.
Campaign volume	Designed for 300+ active campaigns at launch.
Lead data sources	Mixed — both native Meta Lead Forms and landing-page/website leads must be supported.
Campaign naming/tagging	No strict convention exists yet; system ships with a recommended convention plus manual tagging fallback.
Revenue/ROI data	Partial — ROI is estimated using an editable assumed conversion rate and deal value, always labeled "Estimated."
Alert channels	Email + WhatsApp Business API + in-app push (PWA).
Release strategy	MVP first (core tracking, dashboards, alerts); Audience/Creative Analytics and AI-driven features deferred to Phase 2.

29.2 Next Steps

1. Review and mark up this draft (section-by-section comments welcome).
2. Break the MVP (Section 24) into sprint-sized epics for build planning.
3. Confirm WhatsApp Business API account status/lead time before committing to a launch date for that alert channel.
4. ER diagram (Section 18.2): you'll be updating this separately — the entity/relationship list in 18.1 is ready whenever you want to generate it as a visual.